

SUHAIL

DATA SCIENTIST | APPLIED MACHINE LEARNING

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PROFESSIONAL SUMMARY

Data Scientist with 4+ years of experience across data analytics, healthcare, and technology-driven environments, specializing in Python, SQL, predictive modeling, and applied machine learning. Experienced in transforming complex data into analytical and ML solutions, with hands-on work spanning feature engineering, model evaluation, data quality, and reproducible ML workflows. Combines data science expertise with PostgreSQL, REST API, AWS, and production-system experience to build ML solutions designed for real-world integration and deployment.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript

Machine Learning & Data Science: pandas, NumPy, scikit-learn, XGBoost, SciPy, statsmodels, SHAP, feature engineering, predictive modeling, time-series forecasting, model evaluation

MLOps & ML Engineering: MLflow, FastAPI, Docker, GitHub Actions, experiment tracking, model versioning, model serving, monitoring

Data & Visualization: PostgreSQL, Redis, Excel, Power BI, Tableau, Matplotlib

Cloud & Production: AWS (EC2, S3, RDS, IAM), REST APIs, Node.js, Express.js, Nginx, Linux, Git/GitHub

PROFESSIONAL EXPERIENCE

NYRX at Ralph LLC

New York, NY

Data Scientist

Mar 2026 - Present

- Developed a medication demand forecasting solution from pharmacy billing and ordering history, using time-based features and predictive models to estimate future demand and surface potential inventory mismatches.
- Established the ML workflow from data preparation through model evaluation and deployment, incorporating experiment tracking, model versioning, containerized inference, and performance monitoring for repeatable model releases.
- Designed PostgreSQL data models and validation workflows for healthcare and pharmacy data, improving the reliability of data used across analytics, operational reporting, and machine-learning workflows.
- Built and maintained REST APIs and backend services with Node.js, Express.js, PostgreSQL, Redis, and AWS, providing the production infrastructure needed to connect application workflows with analytical and ML components.
- Led backend and data implementation within a five-person engineering team, working directly with company leadership and coordinating with frontend, DevOps, and QA through development, testing, and deployment.

Junior Data Analyst

Oct 2025 - Mar 2026

- Analyzed pharmacy billing, inventory, and operational data using SQL, Excel, and Power BI, cleaning and reconciling records to identify quantity discrepancies, usage patterns, and inventory trends.
- Built reporting workflows and dashboards to compare medication billing and ordering activity, giving management clearer visibility into shortages, excess quantities, and purchasing needs.

Hands Industries FZC

Sharjah, UAE (Remote)

Data Analyst I

Aug 2020 - Oct 2023

- Analyzed customer, sales, and operational datasets using SQL, Excel, Tableau, and Power BI, cleaning and organizing approximately 5,000-10,000 records to identify customer trends, recurring issues, and business performance patterns.
- Built dashboards and structured reporting workflows from manually maintained business records, helping management track customer and operational trends and use data more effectively for decision-making.

PROJECTS

30-Day Hospital Readmission Risk Prediction

Python | pandas | scikit-learn | XGBoost | CatBoost | SHAP

- Built an end-to-end ML pipeline on **100K+ diabetic hospital encounters**, with patient-level leakage prevention, ICD-9 feature engineering, class-imbalance handling, and grouped 5-fold cross-validation.
- Tuned and selected **XGBoost**, achieving **0.232 PR-AUC**, **0.680 ROC-AUC**, and **71.2% recall** on a locked **10K-encounter test set**, with validation-based threshold selection for readmission-risk prioritization.
- Applied **TreeSHAP** and **error analysis** to explain prediction drivers and investigate false positives, false negatives, and subgroup performance disparities.

Pharmacy Demand Forecasting & MLOps Pipeline - NYRX

Python | PostgreSQL | MLflow | FastAPI | Docker | AWS

- Engineered temporal, lag, rolling, and calendar features from pharmacy billing and ordering history and compared forecasting approaches using time-aware validation and baseline-relative evaluation.
- Built a reproducible ML lifecycle covering experiment tracking, model versioning, automated testing, containerized inference, prediction logging, and monitoring for integration with existing application workflows.

RESEARCH & PUBLICATIONS

MedTutorDx — LLM Medical Reasoning & Evaluation Benchmark

Manuscript in Preparation

- Conducting controlled experiments on medical AI reasoning and question-answering across multiple medical subject areas using reproducible evaluation criteria and statistical analysis.

A Desktop-Based Analytics System for Pharmacy Billing and Order Reconciliation

2026 | Technical Report | DOI: 10.5281/zenodo.18652148

- Designed and evaluated an applied pharmacy analytics system for billing and order reconciliation, supporting scalable filtering, discrepancy analysis, and operational decision support across 700+ medications.

EDUCATION

St. Francis College | Brooklyn, NY

Master of Science in Information Technology Management | GPA: 3.85/4.00 | Aug 2025

Osmania University | Hyderabad, India

Bachelor of Engineering in Computer Science & Engineering | GPA: 3.67/4.00 | Jul 2023

LEADERSHIP

President, Ace Lords - Department of Computer Science & Engineering | L.I.E.T | 2021-2023

- Led a 15-member technical organization serving a 300-400 student CSE department; organized 10 technical workshops, a Datathon, and a three-day technical event featuring 8 competitions and 800+ participants. Received a Letter of Appreciation for two years of leadership.